DO NOW — RECALL

Lesson 154 · Search Engines: Crawling, Indexing and Re

Name: _____ Class: _____ Date: _____

1. Why does a server recheck a browser form?

2. When can binary search be used?

3. What data structure follows LIFO?

4. What is a database foreign key?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Client checks are bypassable.

2. Model answer

On sorted data.

3. Model answer

Stack.

4. Model answer

A reference to a key in another table.

By the end, you can…

- Crawler fetches pages found through links and sitemaps.
- Map each searchable term to candidate pages.
- Match query terms and rank relevant results.

Vocabulary — know these terms

crawler — Search agent fetching linked pages.

inverted index — Terms mapped to document identifiers.

tokenisation — Breaking text into searchable units.

ranking — Ordering candidate search results.

NEW LEARNING

Read, annotate and explain

Discover

Crawler fetches pages found through links and sitemaps.

Index

Map each searchable term to candidate pages.

Retrieve

Match query terms and rank relevant results.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

Tokenise both pages

Identify the input or diagram and predict the first result.

Map terms to page IDs

Follow the second stage and explain its importance.

Match search → A

binary→A,B; search→A; tree→B. The query 'binary search' intersects both postings and retrieves A.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain crawling and indexing. [6]

Question 2

Index P='cat dog', Q='dog bird' and identify result for 'dog bird'. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Discuss why a new page might not appear in search and why robots.txt is not a security control. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Search indexing and result ranking are the same operation.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.